

Comparing KNN, NB, and DT

Introduction

In forensic science, the investigation process often requires only a limited amount of evidence left behind at the crime scene. Such evidence includes fragments of glass. Identifying the source of these fragments is crucial in the investigation process. This study is intended to assess and compare the effectiveness of three machine learning models that I intend to develop: K-Nearest Neighbours (KNN), Naïve Bayes (NB), and Decision Trees (DT) on how accurately they can identify the different types of glass.

Data and Preparation

For this research, I am using the glass identification database, which has 214 instances. It has 9 continuous values for the chemical attributes. These include the Refractive Index (RI) and the percentage of certain oxides such as Sodium (Na), Magnesium (Mg), and Silicon (Si). There are 7 target classes, representing different types of glass. Data preparation for the research involved separating the chemical attributes from the types of glass. A 70/30 split was done on the dataset using a randomised permutation with my student ID to ensure reproducibility. As the distance-based algorithms are sensitive to feature magnitudes, the continuous values were standardised using a standard scaler, fit only on the training data to prevent data leakage.

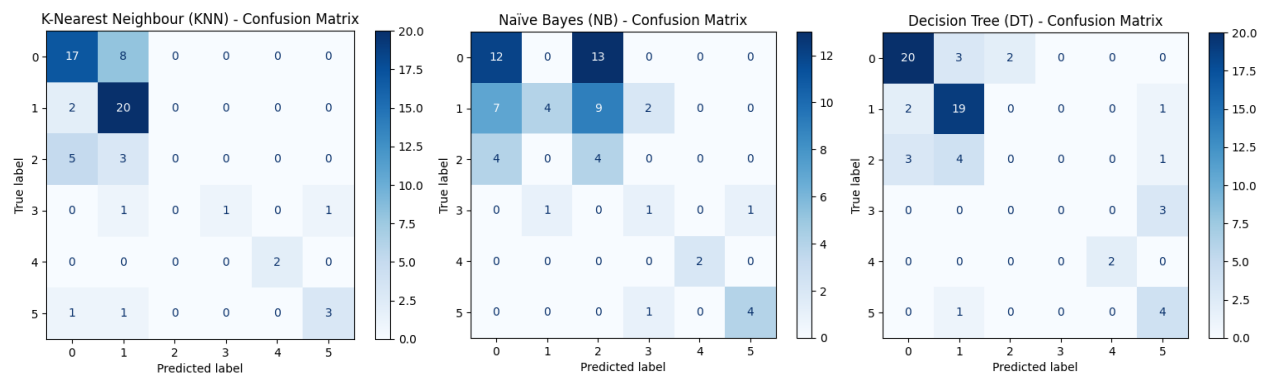
Methodology

Three machine learning algorithms were implemented from scratch to investigate this problem: KNN, DT, and NB. The KNN model was used as a distance-based baseline, NB as a probabilistic baseline, and DT to handle non-linear decision boundaries and to perform implicit feature selection. The models were evaluated using prediction accuracy, training/testing time, and confusion matrices to analyse misclassifications.

Results

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=== Final Results Summary ===
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	Model	Accuracy	Train Time (s)	Test Time (s)
	K-Nearest Neighbour (KNN)	0.661538	0.000000	0.005896
	Naïve Bayes (NB)	0.415385	0.001009	0.012646
	Decision Tree (DT)	0.692308	0.470961	0.000000



The Decision Tree achieved the highest accuracy (69.23%), followed by KNN (66.15%). Timing metrics highlighted the theoretical differences between the algorithms: KNN exhibited near-zero training times but took longer to test, accurately reflecting its nature as a lazy learner that defers computation until testing. Conversely, the Decision Tree is an eager learner; it took the longest to train (~0.36s) to calculate optimal splits but tested instantaneously (0.000s) because predicting merely requires traversing a pre-compiled logical flowchart. While Naïve Bayes is also an eager learner, its training time was effectively zero (~0.001s); computing the Gaussian parameters for a dataset of this small size via vectorized operations is computationally trivial.

The Confusion matrices for DT and KNN showed a distinct diagonal trend of correct predictions. In contrast, the Naïve Bayes matrix exhibited heavy vertical banding. This indicates the model failed to find differentiating patterns and defaulted to frequent classes, impacted by the violation of feature independence and class imbalance.

Conclusion

I highly recommend the Decision Tree model for this classification problem. It achieved the highest accuracy (~69%) and naturally handles correlated chemical features without requiring distance-scaling. Naïve Bayes yielded the lowest accuracy (~41.5%). This is expected, as the algorithm assumes feature independence, a condition heavily violated by the naturally correlated chemical properties of glass. Furthermore, the Gaussian assumption poorly models sparse features like Iron. Future research should consider addressing the severe class imbalance to improve minority class predictions.